



Cyberster

INTERNSHIP REPORT

Week 1: SOC Operations and SIEM Foundations



Submitted to: Cyberster Internship Program

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Cyberster Blue Team Internship — Week 1 Day 1

SOC Foundations: Networking & Traffic Analysis

Today marked the beginning of Phase 1 in the Cyberster Blue Team Internship, focused on building strong networking fundamentals before moving into SIEM deployment and detection engineering.

As emphasized in the task brief Cyberster, understanding how network traffic behaves is critical for any SOC analyst. Instead of memorizing definitions, I focused on tracing real traffic and analyzing it practically.

Understanding Network Models (OSI & TCP/IP)

I reviewed both the OSI and TCP/IP models and mapped how a real web request travels across layers.

To make it practical, I traced what happens when a browser connects to a website:

1. **DNS request (Port 53)** resolves the domain to an IP address
2. **TCP 3-way handshake** establishes connection (SYN → SYN-ACK → ACK)
3. **TLS negotiation** encrypts communication (HTTPS on Port 443)
4. **HTTP request/response exchange** transfers webpage data

This exercise helped me understand how protocols interact across layers rather than viewing them in isolation.

Subnetting & Network Communication

I studied subnetting at a practical level, particularly:

- CIDR notation (e.g., /24)
- How subnet masks define network boundaries
- How to determine whether two IPs can communicate directly

Understanding this is essential for SOC analysis because many alerts involve lateral movement attempts or traffic between unexpected subnets.

Ports & Protocol Awareness for SOC

I reviewed commonly monitored ports in security operations:

- 22 (SSH)
- 80 (HTTP)
- 443 (HTTPS)

- 53 (DNS)
- 445 (SMB)
- 3389 (RDP)
- 25 (SMTP)
- 8080 (Alternate HTTP)

Instead of just memorizing them, I analyzed how abnormal usage (e.g., excessive RDP attempts or suspicious DNS queries) could indicate brute force attempts, malware C2 communication, or lateral movement.

Wireshark Traffic Capture & Analysis

To reinforce theory with practice, I captured 10 minutes of live network traffic using Wireshark and identified multiple protocols including:

- DNS
- TCP
- TLS
- ARP
- ICMP

For each protocol, I reviewed packet details to understand:

- Who initiated communication
- Which ports were used
- Whether traffic appeared normal or suspicious

This was my first step toward thinking like a SOC analyst not just seeing packets, but also interpreting behavior.

Screenshots for each are attached below

Network Diagram

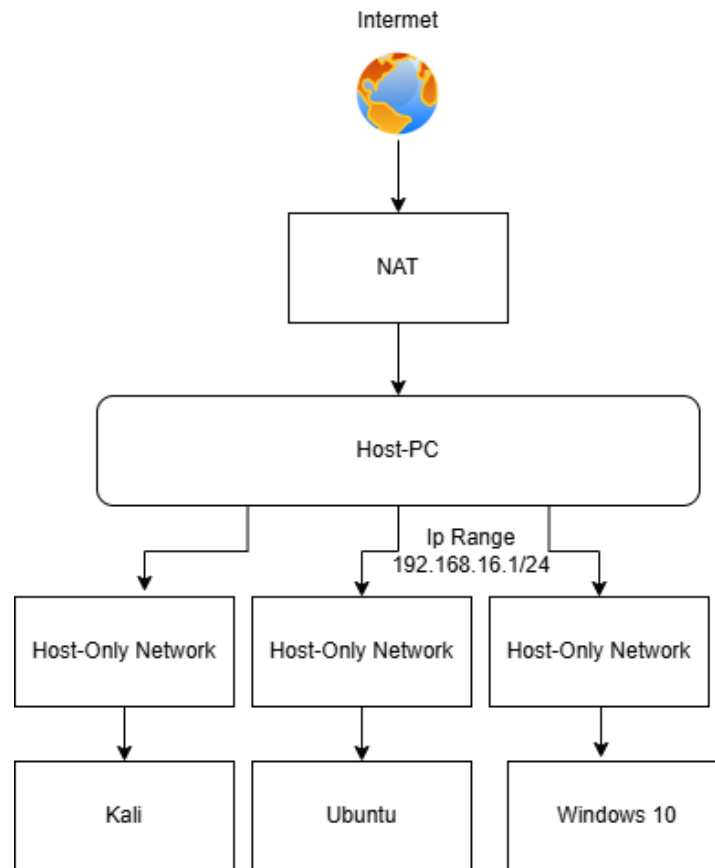


Fig: 1.1 Network Diagram

ICMP Connectivity

I verified network reachability by pinging “8.8.8.8”, observing a consistent sequence of **ICMP Echo Requests and Replies**. This test confirms that my routing and external connectivity are fully functional.

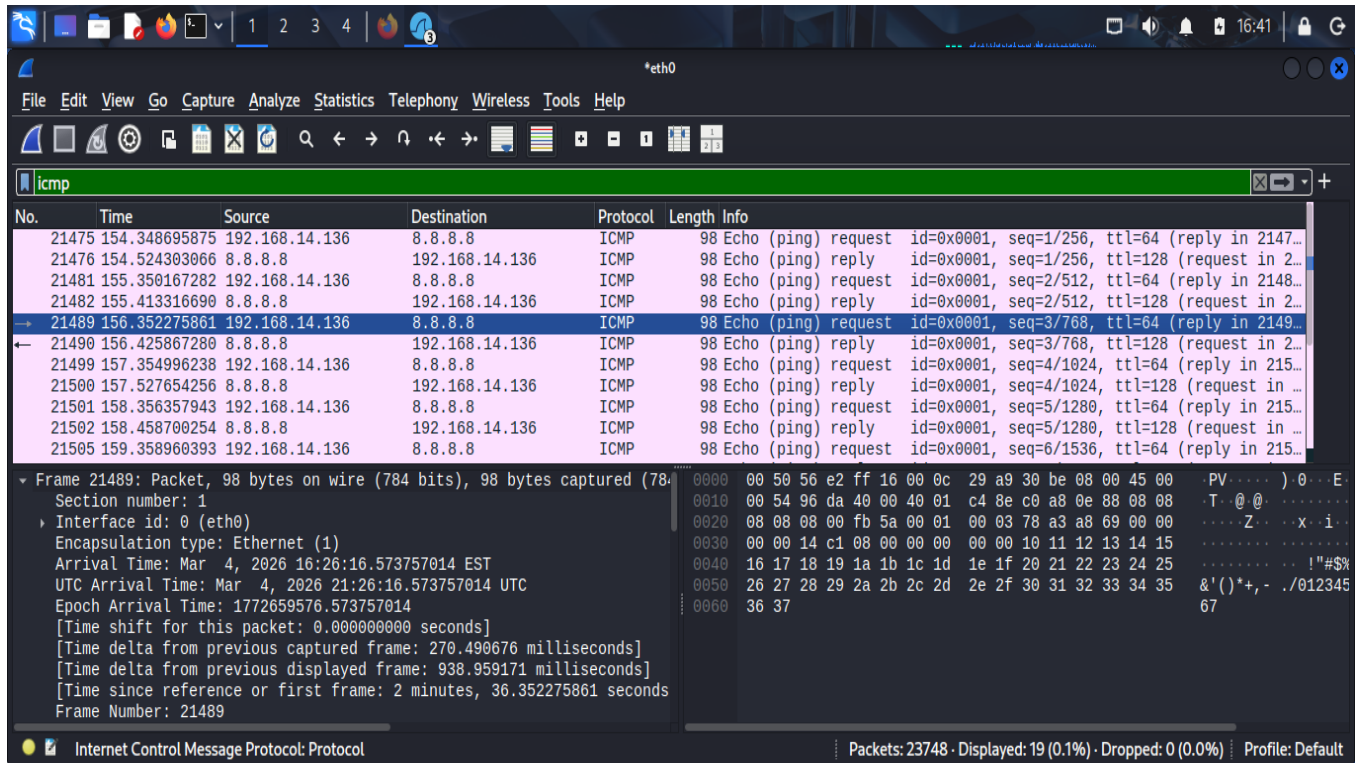


Fig: 1.2 ICMP 8.8.8.8

ARP Broadcasts

I captured **ARP requests** where the source is broadcasting to the entire local network to map an IP address to a MAC address. This is a fundamental Layer 2 process for local hardware communication.

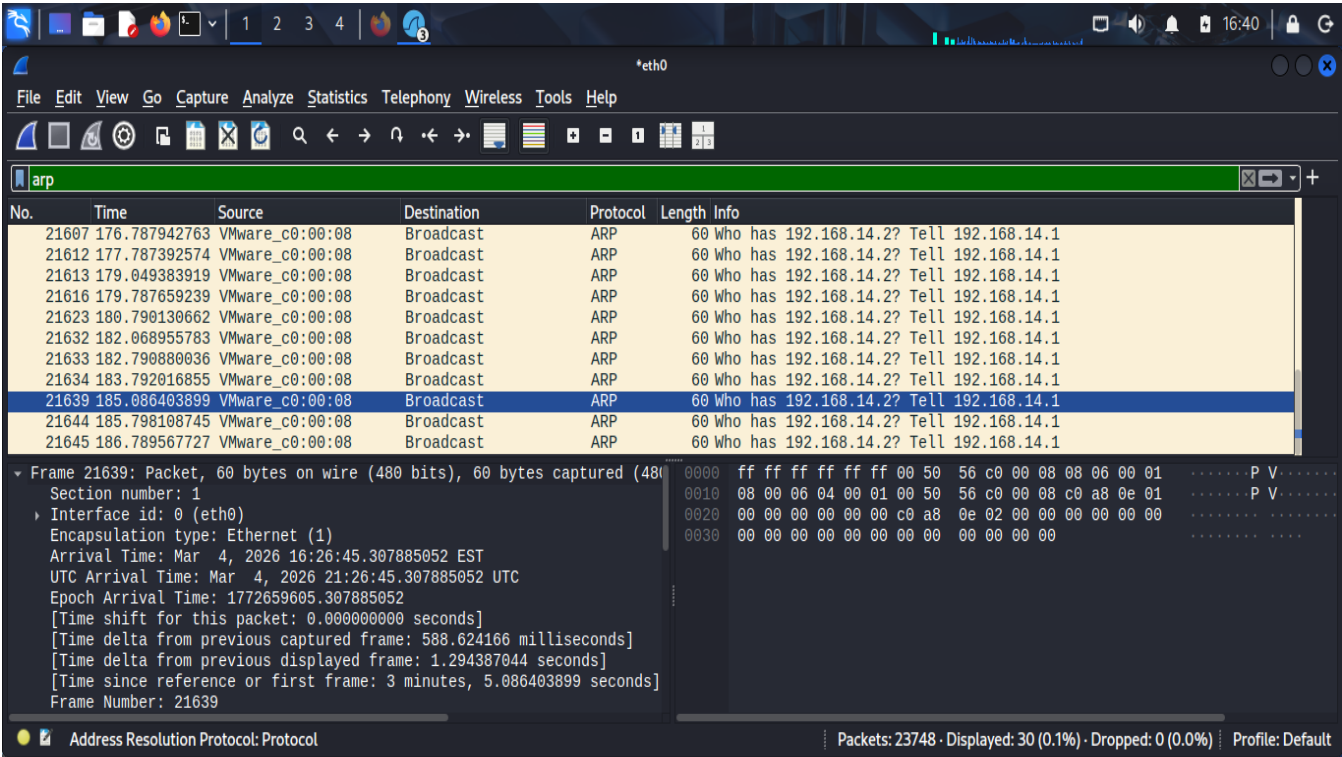


Fig: 1.3 ARP Broadcasts

Secure Session (TLS 1.3)

I monitored the **TLS 1.3** handshake for a Google sub-domain, noting the "Change Cipher Spec" packets. This confirms the successful transition from a plaintext request to a fully encrypted data exchange.

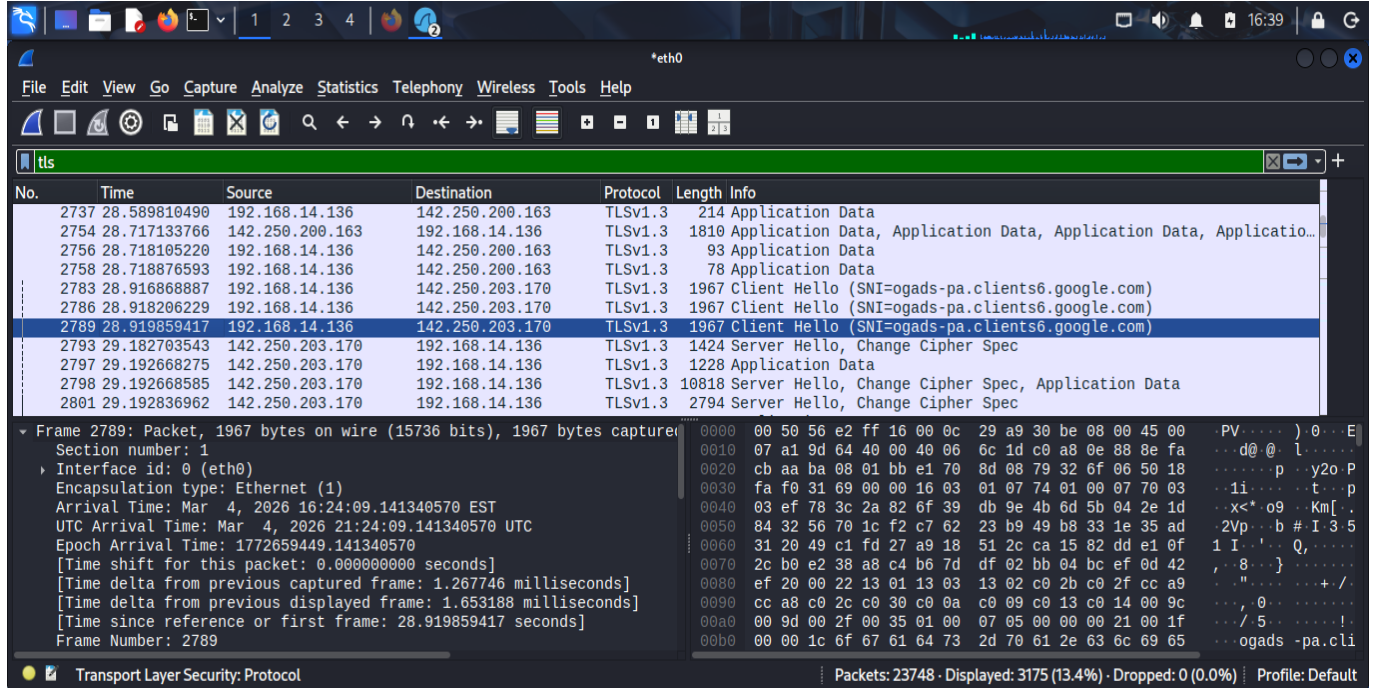


Fig: 1.4 TLS 1.3

TCP Handshake and TLS Initiation

This capture shows the **TCP three-way handshake** followed by a **TLS 1.3 Client Hello**. This identifies the start of a secure connection where my host specifies the target server via SNI.

The image shows a Wireshark packet capture on the interface eth0. The filter is set to 'tcp'. The packet list shows several TCP and TLSv1.3 packets. The packet details pane is expanded for frame 1801, showing the Ethernet II header and the start of the TLSv1.3 Client Hello message. The hex and ASCII views of the packet data are also visible.

No.	Time	Source	Destination	Protocol	Length	Info
1775	25.885624632	142.250.187.35	192.168.14.136	TCP	60	443 → 54258 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
1776	25.885691390	192.168.14.136	142.250.187.35	TCP	54	54258 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
1781	25.887440600	192.168.14.136	142.250.187.35	TLSv1.3	1956	Client Hello (SNI=fonts.gstatic.com)
1782	25.887849471	142.250.187.35	192.168.14.136	TCP	60	443 → 54258 [ACK] Seq=1 Ack=1461 Win=64240 Len=0
1783	25.887849828	142.250.187.35	192.168.14.136	TCP	60	443 → 54258 [ACK] Seq=1 Ack=1903 Win=64240 Len=0
1795	25.892380741	142.250.187.81	192.168.14.136	TCP	60	443 → 49658 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
1796	25.892437608	192.168.14.136	142.250.187.81	TCP	54	49658 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
1801	25.900722887	192.168.14.136	142.250.187.81	TLSv1.3	1957	Client Hello (SNI=csp.withgoogle.com)
1802	25.901170503	142.250.187.81	192.168.14.136	TCP	60	443 → 49658 [ACK] Seq=1 Ack=1461 Win=64240 Len=0
1803	25.901171268	142.250.187.81	192.168.14.136	TCP	60	443 → 49658 [ACK] Seq=1 Ack=1904 Win=64240 Len=0
1827	25.915824146	142.250.187.81	192.168.14.136	TCP	60	443 → 49646 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460

Frame 1801: Packet, 1957 bytes on wire (15656 bits), 1957 bytes captured
Section number: 1
Interface id: 0 (eth0)
Encapsulation type: Ethernet (1)
Arrival Time: Mar 4, 2026 16:24:06.122204040 EST
UTC Arrival Time: Mar 4, 2026 21:24:06.122204040 UTC
Epoch Arrival Time: 1772659446.122204040
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.881357 milliseconds]
[Time delta from previous displayed frame: 8.285279 milliseconds]
[Time since reference or first frame: 25.900722887 seconds]
Frame Number: 1801

0000 00 50 56 e2 ff 16 00 0c 29 a9 30 be 08 00 45 00 PV 0 . . . E
0010 07 97 db cd 40 00 40 06 3e 17 c0 a8 0e 88 8e fa . . . @ . >
0020 bb 51 c1 fa 01 bb cf a3 d7 85 53 a2 6d a3 50 18 Q S m P
0030 fa f0 21 06 00 00 16 03 01 07 6a 01 00 07 66 03 . ! j . . f
0040 03 7b 3e 73 2f 8f 83 4a a9 8a 41 5c fa 40 4e ce { > s / . J . . A \ . @ N
0050 0c 68 a7 7d 31 05 87 28 5c fd 33 cb e1 5a 79 06 h } 1 . (\ 3 . Zy
0060 1c 20 d5 13 86 b5 44 c7 11 ba 44 85 e1 db 1b 69 D . . D . .
0070 37 43 ab 52 cf 00 05 8c b4 10 0d 40 0b ec 46 bf 7C R @ . . F
0080 c5 e1 00 22 13 01 13 03 13 02 c0 2b c0 2f cc a9 . . . " + / .
0090 cc a8 c0 2c c0 30 c0 0a c0 09 c0 13 c0 14 00 9c . . . , 0
00a0 00 9d 00 2f 00 35 01 00 06 fb 00 00 00 17 00 15 . . / 5
00b0 00 00 12 63 73 70 2e 77 69 74 68 67 6f 6f 67 6c . . csp.w ithgoog

Transmission Control Protocol: Protocol Packets: 23748 · Displayed: 9754 (41.1%) · Dropped: 0 (0.0%) Profile: Default

Fig: 1.5 TCP Handshake

DNS Query Analysis

I analyzed a **DNS Type AAAA** request for “**csp.withgoogle.com**”, where my machine specifically asked for the domain's IPv6 address. The hex dump reveals the hostname in plain text before the resolution is completed.

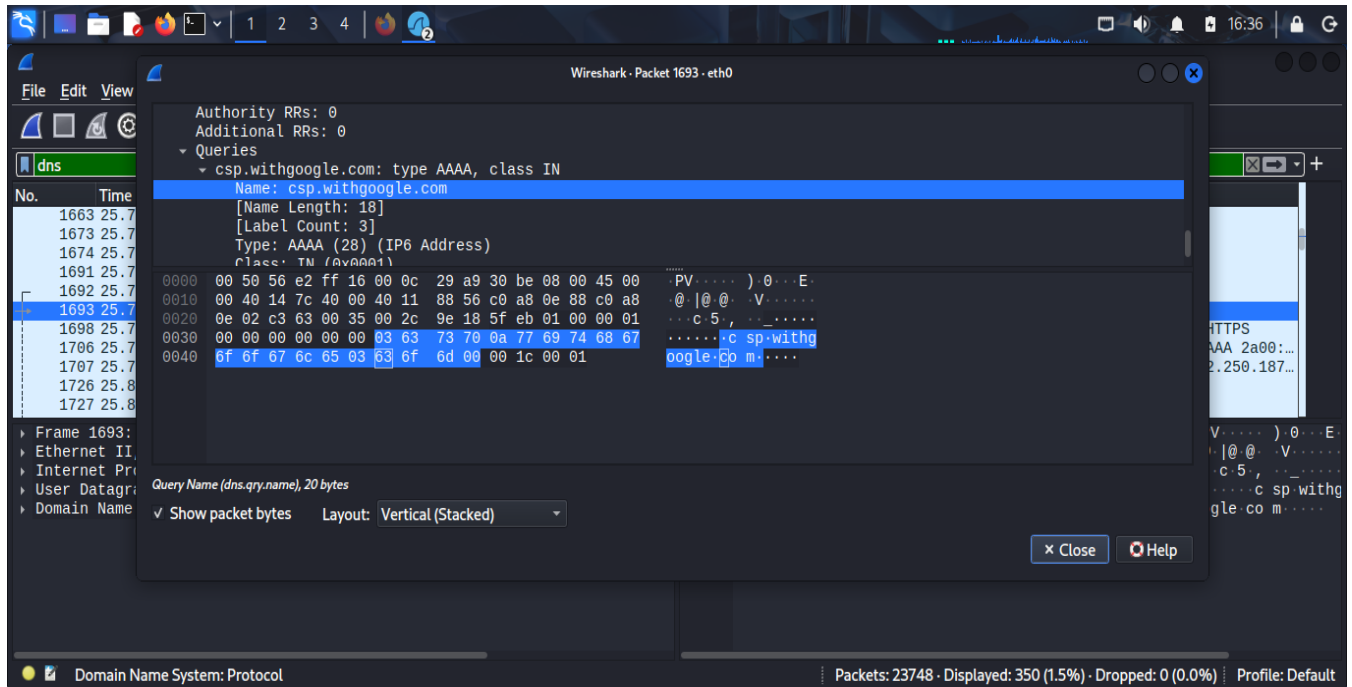


Fig: 1.6 DNS Query

Summary of Day 1

When I am working on a network, everything starts with **DNS**. It is the internet's phonebook, translating a name like "**google.com**" into an IP address so my computer knows who to talk to. Once I have that IP, my machine performs a **TCP Handshake**. Think of this as a quick "**hello**" (**SYN**), a "**hey, I'm here**" (**SYN-ACK**), and a "**got it, let's talk**" (**ACK**). It is a three-step process that makes sure the connection is solid before any real data moves.

Why Structure Matters

We use **subnetting** to keep the network organized. Instead of one giant, noisy room where everyone is shouting, subnetting creates smaller "**offices**." This stops unnecessary traffic from slowing things down and, more importantly, it keeps different parts of the network secure from each other.

The Security Perspective

As a **SOC analyst**, I have to live and breathe this stuff. If I do not know what a normal TCP handshake or a standard DNS query looks like, I will never spot the "**weird**" stuff like an attacker trying to scan the network or hide data in a DNS request. You cannot defend what you do not understand, so knowing the protocols is the only way to actually catch a threat in real-time.

Day 2 – Virtual Lab Environment Setup

Objective

The objective of Day 2 was to create a virtual lab environment using VMware Workstation in order to simulate a SOC monitoring infrastructure. The lab environment consisted of multiple virtual machines including Ubuntu Desktop for Wazuh Manager and Kali Linux for security testing.

Tools Used

- VMware Workstation
- Ubuntu Linux
- Kali Linux
- Windows Host Machine

Procedure

Step 1: Installing VMware Workstation

VMware Workstation was installed on the Windows host machine to create and manage virtual machines.

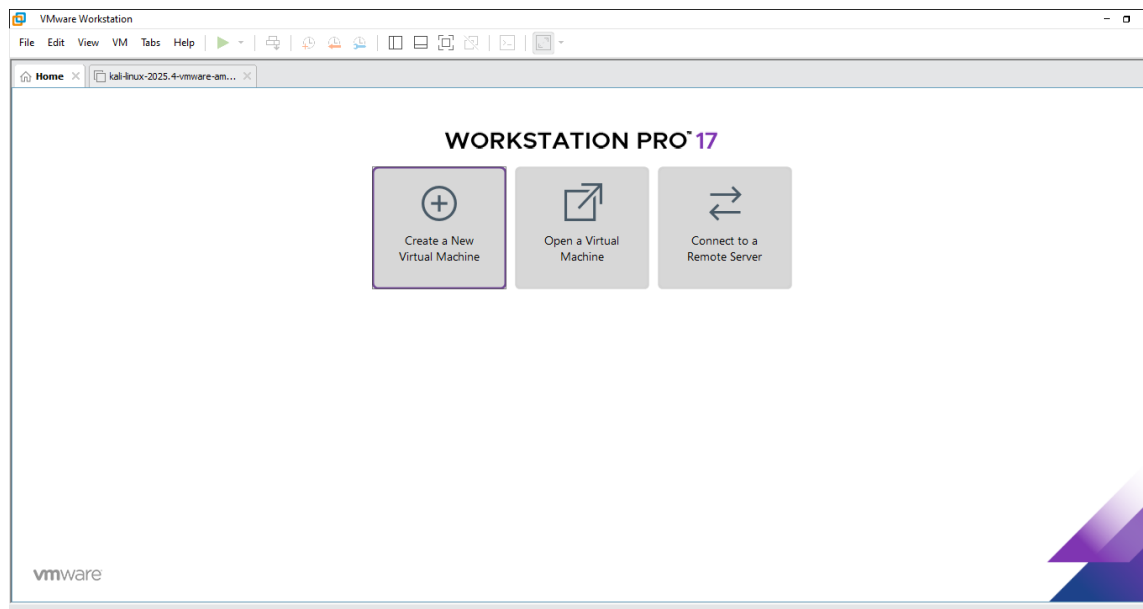


Fig: 2.1 VMware Workstation interface showing the virtual machine environment

Step 2: Creating Ubuntu Virtual Machine

A new virtual machine was created for Ubuntu which will later host the Wazuh Manager.

The following configuration was used:

- RAM: 4 GB
- CPU: 2 Processors
- Disk Space: 40 GB
- Network Adapter: NAT / Host-only

Ubuntu was then installed successfully inside VMware.

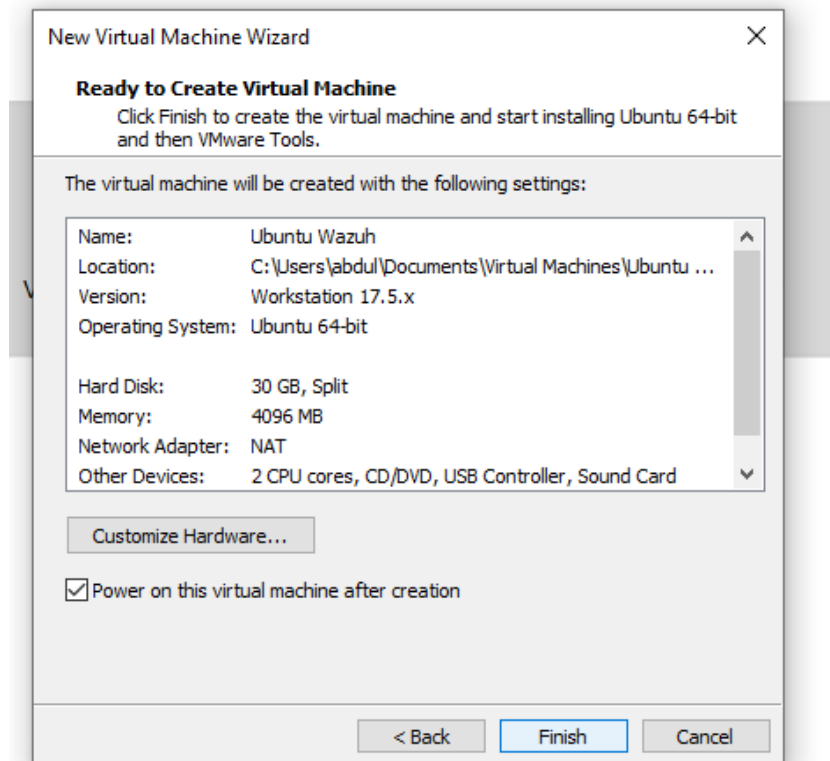


Fig: 2.2 Ubuntu virtual machine configuration in VMware Workstation

Step 3: Creating Kali Linux Virtual Machine

I had already created the virtual machine for Kali Linux. This machine will act as an endpoint that will later send logs to the Wazuh manager.

Configuration used:

- RAM: 2 GB
- CPU: 2 Processors

- Disk: 30 GB

Step 4: Verifying Virtual Machines

Both virtual machines were powered on and tested to ensure they were functioning properly.

Result

The virtual lab environment was successfully created. Ubuntu and Kali Linux virtual machines were installed and running properly inside VMware Workstation.

Day 3 – Installation of Wazuh Manager

Objective

The objective of Day 3 was to install the Wazuh SIEM platform on the Ubuntu virtual machine in order to monitor logs from different endpoints.

Tools Used

- Ubuntu Server
- Wazuh Installation Script
- Terminal

Procedure

Step 1: Updating Ubuntu System

The Ubuntu system was updated to ensure all packages were up to date.

```
sudo apt update
sudo apt upgrade -y
```

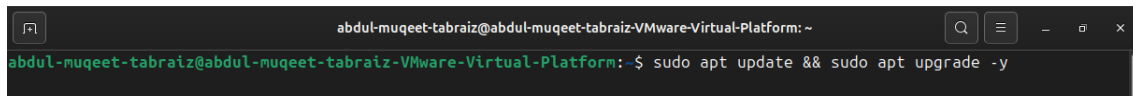
A terminal window with a dark background. The title bar reads 'abdul-muqeeet-tabraiz@abdul-muqeeet-tabraiz-VMware-Virtual-Platform: ~'. The prompt is 'abdul-muqeeet-tabraiz@abdul-muqeeet-tabraiz-VMware-Virtual-Platform:~\$'. The command 'sudo apt update && sudo apt upgrade -y' has been entered and executed, with the output 'Hit:1 http://archive.ubuntu.com/ubuntu InRelease' visible on the next line.

Fig: 3.1 Updating Ubuntu system packages using apt update and upgrade commands

Step 2: Downloading Wazuh Installation Script

The Wazuh installation script was downloaded from the official repository.

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
```

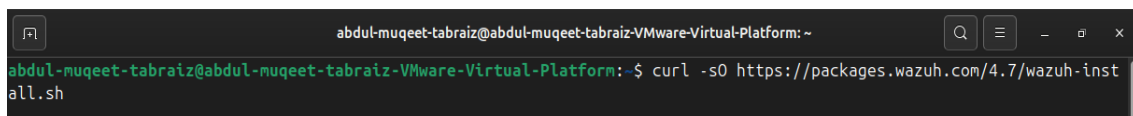
A terminal window with a dark background. The title bar reads 'abdul-muqeeet-tabraiz@abdul-muqeeet-tabraiz-VMware-Virtual-Platform: ~'. The prompt is 'abdul-muqeeet-tabraiz@abdul-muqeeet-tabraiz-VMware-Virtual-Platform:~\$'. The command 'curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh' has been entered and executed, with the output 'all.sh' visible on the next line.

Fig: 3.2 Downloading the Wazuh installation script from the official repository

Step 3: Making Script Executable

```
chmod +x wazuh-install.sh
```

A terminal window with a dark background. The prompt is 'abdul-muqeeet-tabraiz@abdul-muqeeet-tabraiz-VMware-Virtual-Platform:'. The command 'chmod +x wazuh-install.sh' has been entered and executed.

```
abdul-muqeeet-tabraiz@abdul-muqeeet-tabraiz-VMware-Virtual-Platform:~$ chmod +x wazuh-install.sh
```

Fig: 3.3 chmod to change the mod of installation

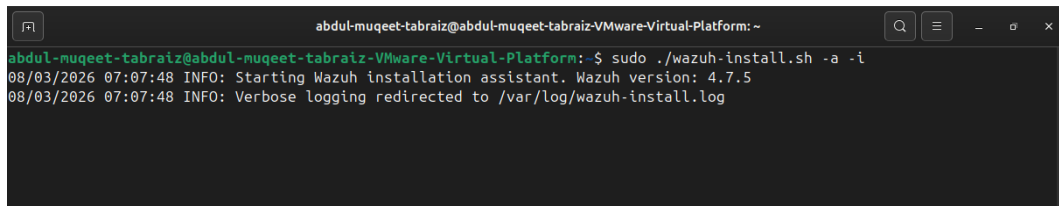
Step 4: Installing Wazuh

The installation assistant was executed.

```
sudo ./wazuh-install.sh -a -i
```

This command installed:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

A terminal window showing the execution of the Wazuh installation script. The prompt is 'abdul-muqeeet-tabraiz@abdul-muqeeet-tabraiz-VMware-Virtual-Platform:'. The command 'sudo ./wazuh-install.sh -a -i' has been entered. The output shows two INFO messages: 'Starting Wazuh installation assistant. Wazuh version: 4.7.5' and 'Verbose logging redirected to /var/log/wazuh-install.log'.

```
abdul-muqeeet-tabraiz@abdul-muqeeet-tabraiz-VMware-Virtual-Platform:~$ sudo ./wazuh-install.sh -a -i
08/03/2026 07:07:48 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
08/03/2026 07:07:48 INFO: Verbose logging redirected to /var/log/wazuh-install.log
```

Fig: 3.4 Wazuh installation process running on the Ubuntu Server

```
abdul-muqheet-tabraiz@abdul-muqheet-tabraiz-VMware-Virtual-Platform: ~
08/03/2026 07:21:43 INFO: Wazuh indexer installation finished.
08/03/2026 07:21:44 INFO: Wazuh indexer post-install configuration finished.
08/03/2026 07:21:44 INFO: Starting service wazuh-indexer.
08/03/2026 07:23:12 INFO: wazuh-indexer service started.
08/03/2026 07:23:12 INFO: Initializing Wazuh indexer cluster security settings.
08/03/2026 07:23:40 INFO: Wazuh indexer cluster initialized.
08/03/2026 07:23:40 INFO: --- Wazuh server ---
08/03/2026 07:23:40 INFO: Starting the Wazuh manager installation.
08/03/2026 07:29:45 INFO: Wazuh manager installation finished.
08/03/2026 07:29:45 INFO: Starting service wazuh-manager.
08/03/2026 07:30:20 INFO: wazuh-manager service started.
08/03/2026 07:30:20 INFO: Starting Filebeat installation.
08/03/2026 07:31:09 INFO: Filebeat installation finished.
08/03/2026 07:31:12 INFO: Filebeat post-install configuration finished.
08/03/2026 07:31:12 INFO: Starting service filebeat.
08/03/2026 07:31:23 INFO: filebeat service started.
08/03/2026 07:31:23 INFO: --- Wazuh dashboard ---
08/03/2026 07:31:24 INFO: Starting Wazuh dashboard installation.
08/03/2026 07:36:40 INFO: Wazuh dashboard installation finished.
08/03/2026 07:36:40 INFO: Wazuh dashboard post-install configuration finished.
08/03/2026 07:36:40 INFO: Starting service wazuh-dashboard.
08/03/2026 07:36:43 INFO: wazuh-dashboard service started.
08/03/2026 07:38:10 INFO: Initializing Wazuh dashboard web application.
08/03/2026 07:38:12 INFO: Wazuh dashboard web application initialized.
08/03/2026 07:38:12 INFO: --- Summary ---
08/03/2026 07:38:12 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password:
08/03/2026 07:38:12 INFO: --- Dependencies ---
08/03/2026 07:38:12 INFO: Removing gawk.
08/03/2026 07:38:25 INFO: Installation finished.
abdul-muqheet-tabraiz@abdul-muqheet-tabraiz-VMware-Virtual-Platform:~$
```

Fig: 3.5 Successful installation of Wazuh Manager, Indexer, and Dashboard components

Result

The Wazuh platform was successfully installed on the Ubuntu virtual machine and all components were deployed.

Day 4 – Accessing the Wazuh Dashboard

Objective

The objective of Day 4 was to access the Wazuh dashboard through the browser and verify that the SIEM system was operational.

Procedure

Step 1: Finding Ubuntu IP Address

The IP address of the Ubuntu machine was obtained using the following command:

```
ip a
```

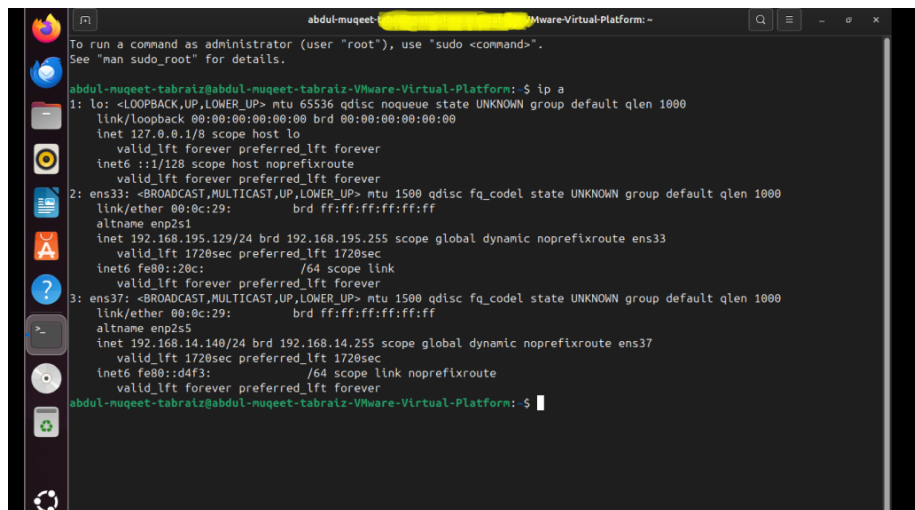
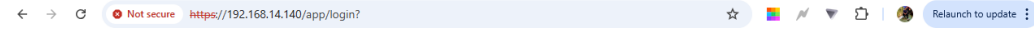


Fig: 4.1 Ubuntu system displaying IP address using the ip a command

Step 2: Accessing the Dashboard

The Wazuh dashboard was accessed using a web browser.

```
https://192.168.14.140
```



wazuh.

Loading ...

Fig: 4.2 Accessing the Wazuh Dashboard through a web browser

Step 3: Logging into Dashboard

Login credentials generated during installation were used.

```
Username: admin  
Password: *****
```

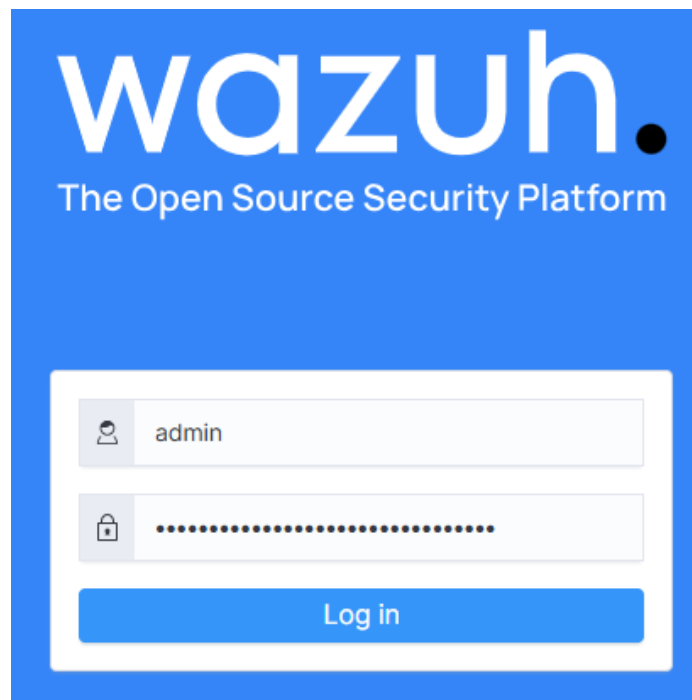


Fig: 4.3 Wazuh Dashboard Login Page

Step 4: Exploring Dashboard

The dashboard interface was explored to understand:

- Security Events
- Agents
- Alerts
- Monitoring Overview

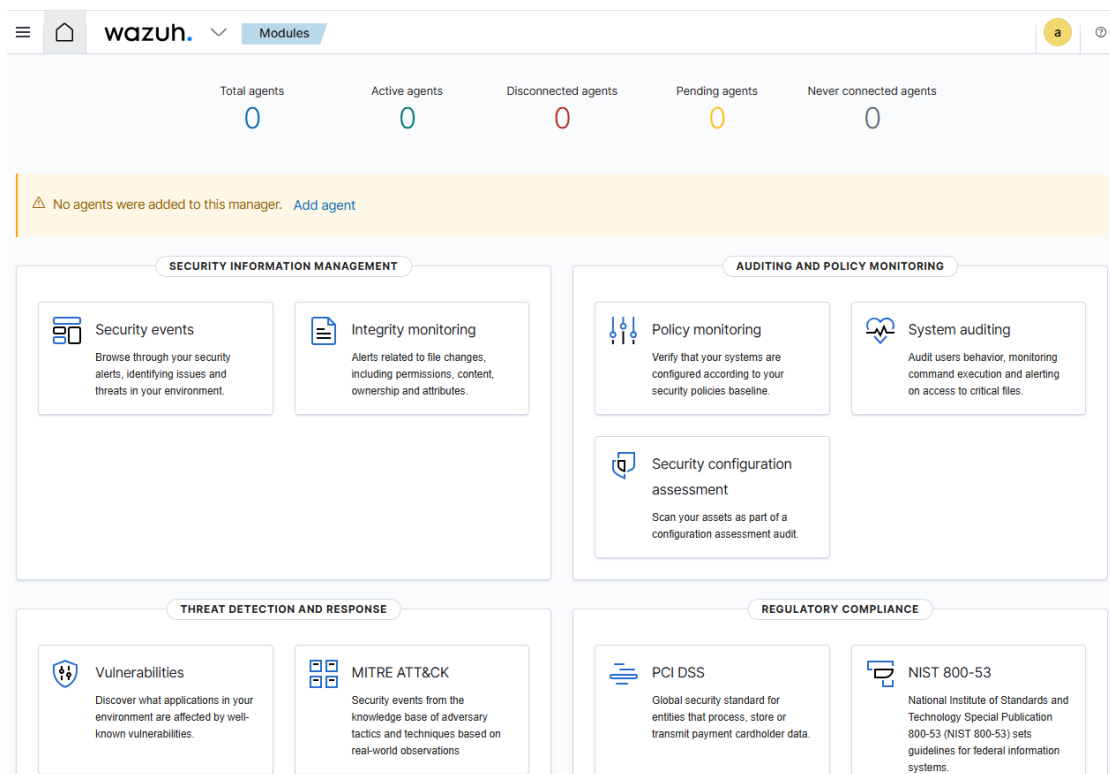


Fig: 4.4 Wazuh Dashboard showing security monitoring interface

Result

The Wazuh dashboard was successfully accessed and verified that the SIEM system was working correctly.

Day 5 – Wazuh Agent Installation on Kali Linux

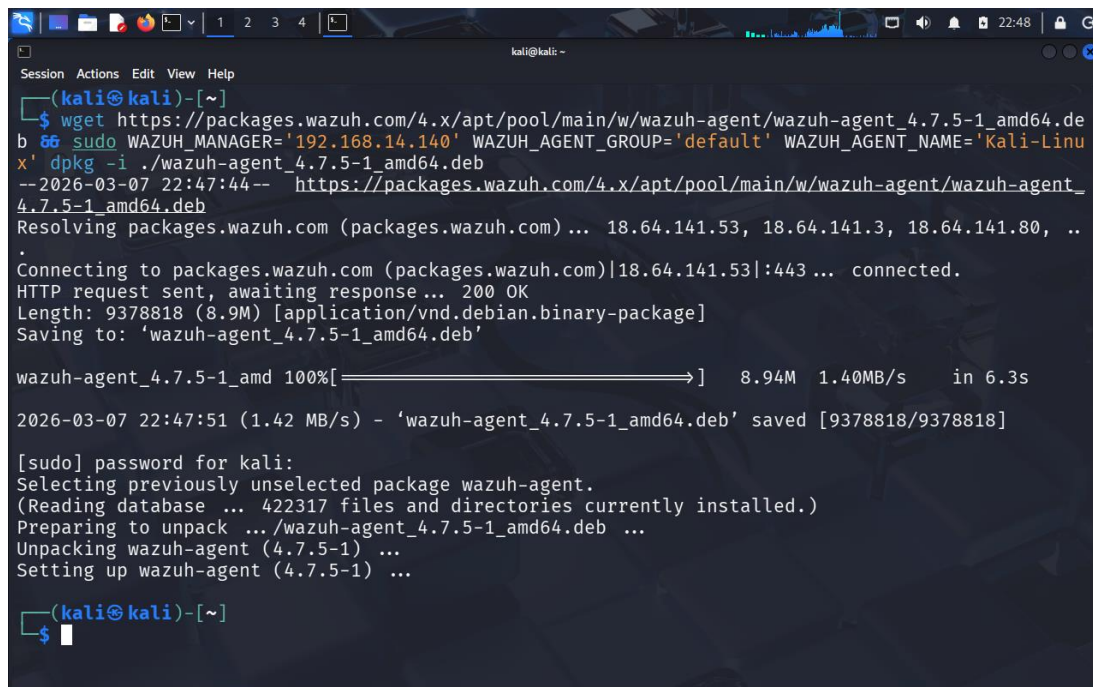
Objective

The objective of Day 5 was to install the Wazuh agent on Kali Linux so that logs from the Kali machine could be monitored by the Wazuh manager.

Procedure

Step 1: Adding Wazuh Repository

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo apt-key add -
```



```
kali@kali: ~  
$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb  
b 66 sudo WAZUH_MANAGER='192.168.14.140' WAZUH_AGENT_GROUP='default' WAZUH_AGENT_NAME='Kali-Linu  
x' dpkg -i ./wazuh-agent_4.7.5-1_amd64.deb  
--2026-03-07 22:47:44-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_  
4.7.5-1_amd64.deb  
Resolving packages.wazuh.com (packages.wazuh.com)... 18.64.141.53, 18.64.141.3, 18.64.141.80, ..  
.  
Connecting to packages.wazuh.com (packages.wazuh.com)|18.64.141.53|:443 ... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 9378818 (8.9M) [application/vnd.debian.binary-package]  
Saving to: 'wazuh-agent_4.7.5-1_amd64.deb'  
  
wazuh-agent_4.7.5-1_amd 100%[=====>] 8.94M 1.40MB/s in 6.3s  
  
2026-03-07 22:47:51 (1.42 MB/s) - 'wazuh-agent_4.7.5-1_amd64.deb' saved [9378818/9378818]  
  
[sudo] password for kali:  
Selecting previously unselected package wazuh-agent.  
(Reading database ... 422317 files and directories currently installed.)  
Preparing to unpack .../wazuh-agent_4.7.5-1_amd64.deb ...  
Unpacking wazuh-agent (4.7.5-1) ...  
Setting up wazuh-agent (4.7.5-1) ...  
  
(kali@kali)~$
```

Fig: 5.1 Adding Wazuh repository to Kali Linux

Step 2: Installing Wazuh Agent

```
sudo apt install wazuh-agent
```

Step 3: Configuring Manager IP

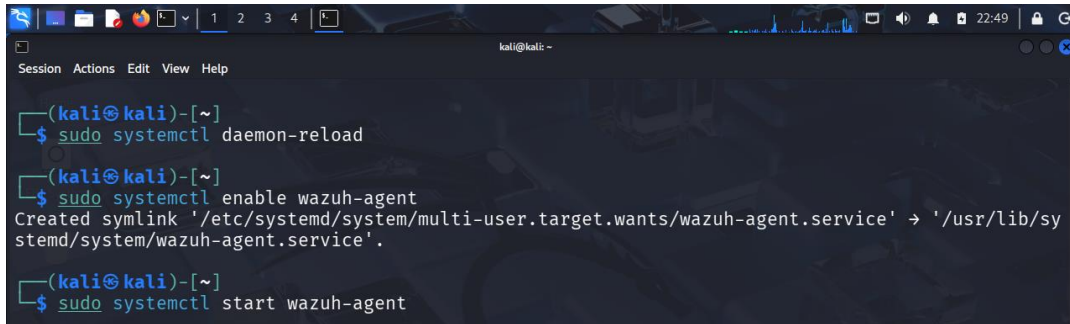
The agent configuration file was edited.

```
sudo nano /var/ossec/etc/ossec.conf
```

Manager IP was added.

Step 4: Starting Agent

```
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent
```

A terminal window titled 'kali@kali: ~' showing the execution of three systemctl commands. The first command is 'sudo systemctl daemon-reload'. The second command is 'sudo systemctl enable wazuh-agent', which outputs 'Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' -> '/usr/lib/systemd/system/wazuh-agent.service''. The third command is 'sudo systemctl start wazuh-agent'.

```
(kali@kali)-[~]  
$ sudo systemctl daemon-reload  
  
(kali@kali)-[~]  
$ sudo systemctl enable wazuh-agent  
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' -> '/usr/lib/systemd/system/wazuh-agent.service'.  
  
(kali@kali)-[~]  
$ sudo systemctl start wazuh-agent
```

Fig: 5.2 Starting and enabling the Wazuh agent service on Kali Linux

Result

Kali Linux was successfully connected to the Wazuh manager as an endpoint agent.

Day 6 – Wazuh Agent Installation on Windows

Objective

The objective of Day 6 was to install the Wazuh agent on the Windows host machine to collect security logs from Windows systems.

Procedure

Step 1: Downloading Windows Agent

The Wazuh Windows agent installer was downloaded from the official website.

Step 2: Installing Agent via PowerShell

PowerShell was opened with administrator privileges and the installation command was executed.

```
msiexec.exe /i wazuh-agent.msi /q WAZUH_MANAGER="192.168.14.140"  
WAZUH_AGENT_NAME="windows-agent"
```

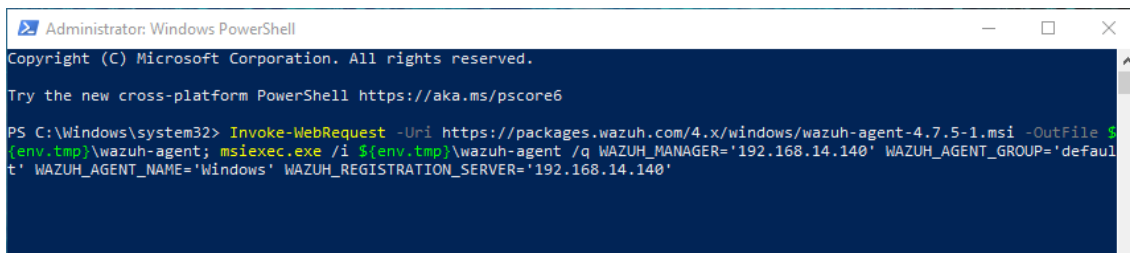


Fig: 6.1 Installing the Wazuh windows agent using PowerShell

Step 3: Starting Wazuh Service

```
NET START WazuhSvc
```

```
Administrator: Windows PowerShell
PS C:\Windows\system32> NET START WazuhSvc
The Wazuh service is starting.
The Wazuh service was started successfully.

PS C:\Windows\system32>
```

Fig: 6.2 Starting the Wazuh agent service on Windows

Step 4: Verifying Agent

The agent appeared in the Wazuh dashboard under the Agents section.

Agents (2) Deploy new agent Refresh Export formatted WQL Refresh

ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
001	Kali-Linux	192.168.14.139	default	Kali GNU/Linux 2025.4	node01	v4.7.5	active	
002	Windows	192.168.14.1	default	Microsoft Windows 10 Pro 10.0.19045.6466	node01	v4.7.5	active	

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Fig: 6.3 Windows agent successfully connected and visible in Wazuh dashboard

Result

The Windows system was successfully integrated into the Wazuh monitoring infrastructure.

Day 7 – Security Event Monitoring

Objective

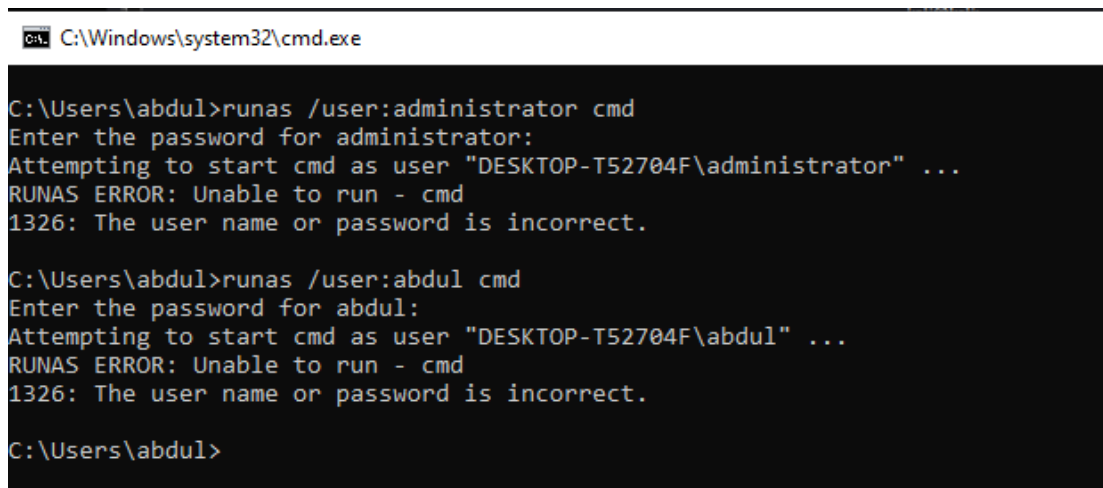
The objective of Day 7 was to monitor security alerts generated by the connected endpoints and understand how a SOC analyst investigates events.

Procedure

Step 1: Generating Test Events

Security related commands were executed on Kali and Windows to generate logs such as:

- Failed login attempts
- User commands
- Authentication events



```
C:\Windows\system32\cmd.exe

C:\Users\abdul>runas /user:administrator cmd
Enter the password for administrator:
Attempting to start cmd as user "DESKTOP-T52704F\administrator" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\abdul>runas /user:abdul cmd
Enter the password for abdul:
Attempting to start cmd as user "DESKTOP-T52704F\abdul" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\abdul>
```

Fig: 7.1 Generating test security events on Windows system

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)~  
$ sudo su clay  
[sudo] password for kali:  
Sorry, try again.  
[sudo] password for kali:  
Sorry, try again.  
[sudo] password for kali:  
sudo: 3 incorrect password attempts  
(kali@kali)~  
$ su root  
Password:  
su: Authentication failure  
(kali@kali)~  
$ sudo su kali  
[sudo] password for kali:  
Sorry, try again.  
[sudo] password for kali:  
(kali@kali)~  
$
```

Fig: 7.2 Generating test security events on Kali Linux

Step 2: Monitoring Alerts

Alerts were observed in the Wazuh dashboard under:

Security Events

Security Alerts						
Time ↓	Technique(s)		Tactic(s)	Description	Level	Rule ID
> Mar 8, 2026 @ 09:28:31.657	T1078	T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
> Mar 8, 2026 @ 09:26:34.376	T1078	T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
> Mar 8, 2026 @ 09:25:35.010	T1078	T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122

Fig: 7.3 Security events detected and displayed in the Wazuh dashboard

Step 3: Analyzing Event Details

Each alert was inspected to view:

- Rule ID
- Severity Level
- Agent Name
- Event Description
- Timestamp

Step 4: Log Investigation

Logs were filtered by agent name to analyze events from specific machines.

Result

Security events generated by Kali Linux and Windows machines were successfully detected and analyzed using the Wazuh dashboard.

Conclusion

During this week, a complete SOC lab environment was successfully deployed using VMware Workstation. The Wazuh SIEM platform was installed and configured on Ubuntu, and agents were deployed on Kali Linux and Windows systems.

Logs from multiple endpoints were collected and analyzed through the Wazuh dashboard, providing practical experience in security monitoring, alert detection, and incident analysis. This hands-on practice helped in understanding the workflow of a Security Operations Center (SOC) analyst.